

Lab sheet-4

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```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Set overall style
```

```
sns.set_theme(style="whitegrid")
```

```
plt.rcParams.update({'font.sans-serif': 'DejaVu Sans', 'font.size': 10})
```

```
# Step 1 & 2: Load the Dataset and Inspect
```

```
filename = "netflix_titles.csv"
```

```
df = pd.read_csv(filename)
```

```
print(f"Dataset successfully loaded. Shape: {df.shape}\n")
```

```
# Preprocessing: Extract numeric movie duration and release year
```

```
df['date_added'] = pd.to_datetime(df['date_added'].str.strip(), errors='coerce')
```

```
df['year_added'] = df['date_added'].dt.year
```

```
# Extract duration integer for movies
```

```
movies_df = df[df['type'] == 'Movie'].copy()
```

```
movies_df['duration_min'] = movies_df['duration'].str.extract('(\d+)').astype(float)
```

```
# Step 3: Bar Chart - Content Type Distribution (Movies vs TV Shows)
```

```
plt.figure(figsize=(7, 5))
```

```
type_counts = df['type'].value_counts()
```

```
sns.barplot(x=type_counts.index, y=type_counts.values, palette='Blues_d')
```

```
plt.title("Distribution of Content Type on Netflix", fontsize=14, weight='bold')
```

```
plt.xlabel("Content Type", fontsize=12)
```

```
plt.ylabel("Total Count", fontsize=12)
```

```
plt.savefig("1_bar_chart_content_type.png", dpi=300, bbox_inches='tight')
```

```
plt.close()
```

```
print("[SAVED] 1_bar_chart_content_type.png")
```

Step 4: Line Chart - Releases Added Per Year Over Time

```
plt.figure(figsize=(10, 5))

yearly_trend = df.groupby(['year_added', 'type']).size().unstack().fillna(0)

# Filter realistic release years for Netflix catalog additions
yearly_trend = yearly_trend[yearly_trend.index >= 2010]

plt.plot(yearly_trend.index, yearly_trend['Movie'], marker='o', label='Movies',
color='#e50914', linewidth=2)

plt.plot(yearly_trend.index, yearly_trend['TV Show'], marker='s', label='TV Shows',
color='#221f1f', linewidth=2)

plt.title("Netflix Catalog Growth Over Time (2010 - Present)", fontsize=14, weight='bold')

plt.xlabel("Year Added", fontsize=12)

plt.ylabel("Number of Titles Added", fontsize=12)

plt.legend()

plt.savefig("2_line_chart_yearly_trend.png", dpi=300, bbox_inches='tight')

plt.close()

print("[SAVED] 2_line_chart_yearly_trend.png")
```

Step 5: Histogram - Distribution of Movie Durations (in Minutes)

```
plt.figure(figsize=(8, 5))

sns.histplot(movies_df['duration_min'].dropna(), bins=30, kde=True, color='#e50914')

plt.title("Distribution of Movie Durations on Netflix", fontsize=14, weight='bold')

plt.xlabel("Duration (Minutes)", fontsize=12)

plt.ylabel("Number of Movies", fontsize=12)

plt.savefig("3_histogram_movie_duration.png", dpi=300, bbox_inches='tight')

plt.close()

print("[SAVED] 3_histogram_movie_duration.png")
```

Step 6: Box Plot - Movie Durations across Top Ratings

```
plt.figure(figsize=(9, 5))

top_ratings = movies_df['rating'].value_counts().index[:6]

filtered_ratings = movies_df[movies_df['rating'].isin(top_ratings)]
```

```

sns.boxplot(data=filtered_ratings, x='rating', y='duration_min', palette='Set2')

plt.title("Movie Duration Distribution Across Top Content Ratings", fontsize=14,
weight='bold')

plt.xlabel("Content Rating", fontsize=12)

plt.ylabel("Duration (Minutes)", fontsize=12)

plt.savefig("4_boxplot_duration_by_rating.png", dpi=300, bbox_inches='tight')

plt.close()

print("[SAVED] 4_boxplot_duration_by_rating.png")

# Step 7: Scatter Plot - Release Year vs Movie Duration

plt.figure(figsize=(9, 5))

sns.scatterplot(

    data=movies_df[movies_df['release_year'] >= 1990],

    x='release_year',

    y='duration_min',

    alpha=0.4,

    color='#e50914'

)

plt.title("Release Year vs. Movie Duration (1990 - Present)", fontsize=14, weight='bold')

plt.xlabel("Release Year", fontsize=12)

plt.ylabel("Duration (Minutes)", fontsize=12)

plt.savefig("5_scatterplot_year_vs_duration.png", dpi=300, bbox_inches='tight')

plt.close()

print("[SAVED] 5_scatterplot_year_vs_duration.png")

# Step 8: Correlation Heatmap of Numerical Features

plt.figure(figsize=(7, 5))

num_df = df[['release_year']].copy()

num_df['year_added'] = df['year_added']

```

```

num_df['duration_min'] = movies_df['duration_min']

corr = num_df.corr(method='pearson')

sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", vmin=-1, vmax=1)

plt.title("Correlation Matrix of Numeric Netflix Attributes", fontsize=14, weight='bold')

plt.savefig("6_heatmap_correlation.png", dpi=300, bbox_inches='tight')

plt.close()

print("[SAVED] 6_heatmap_correlation.png")

```



